BY ORDER OF THE SECRETARY OF THE AIR FORCE

DEPARTMENT OF THE AIR FORCE INSTRUCTION 10-2601

18 SEPTEMBER 2025

Operations

ELECTROMAGNETIC PULSE SURVIVABILITY PROGRAM

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available on the e-Publishing website at www.e-Publishing.af.mil for downloading or ordering

RELEASABILITY: There are no restrictions on the release of this publication

---

OPR: AF/A10XC, SF COO/S10

Certified by: AF/A10
(Lt Gen Andrew J. Gebara)

Service Authority: SF COO/ S10
(Brig Gen Nathan D. Yates)

Supersedes: DAFI10-2601, 11 January 2022

Pages: 36

This Department of the Air Force (DAF) Instruction (DAFI) implements Air Force Policy Directive (AFPD) 10-9, *Lead Command/Lead Agent Designation and responsibilities for United States Weapon Systems, Non-Weapon Systems, and Activities*; and AFPD 10-26, *Countering Weapons of Mass Destruction*. It is consistent with AFPD 10-24, *Mission Assurance*; AFPD 13-5, *Air Force Nuclear Mission*; Air Force Instruction (AFI) 10-2607, *Chemical, Biological, Radiological, and Nuclear Survivability*; AFI 13-500, *Air Force Nuclear Mission Responsibilities*; and DAFI 20-101_63-101, *Integrated Life Cycle Management*. This instruction establishes the DAF Electromagnetic Pulse (EMP) Survivability Program and the roles and responsibilities to ensure it is integrated into the planning, programming, budgeting, and execution process. This publication applies to all civilian employees and uniformed members of the Regular Air Force, the Air Force Reserve, the Air National Guard, the United States Space Force (USSF), and those with a contractual obligation to abide by the terms of DAF publications. It applies to personnel who develop, manage or operate conventional or nuclear mission-critical systems and/or infrastructure requiring EMP survivability in order to operate in and through, or quickly recover from an EMP event. For the purposes of this Instruction, reference to Major Commands (MAJCOM)/Field Commands (FLDCOM) should include Direct Reporting Units (DRUs) when applicable. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the

Office of Primary Responsibility (OPR) using DAF Form 847, *Recommendation for Change of Publication*; route DAF Form 847 from the field through the appropriate functional chain of command. The authorities to waive wing/unit-level requirements will be identified by tier number ("T-0, T-1, T-2, T-3") following the compliance statement. See DAF Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the tier numbers. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority, or alternately, to the requestor's commander for non-tiered compliance items. This publication may be supplemented at any level, but all supplements must be routed to the OPR of this publication for coordination prior to certification and approval.

SUMMARY OF CHANGES

This document has been revised and should be completely reviewed. Changes include updating EMP Test Priority Lists (ETPL) from equipment, facility, aircraft lists to mission-centric lists. Additionally, changes to USSF structure are included. Furthermore, roles/responsibilities pertaining to the Nuclear Oversight Council are defined.

Chapter 1—INTRODUCTION **4**

1.1. Background. 4

1.2. Implementation. 4

1.3. Objectives. 5

Chapter 2—ROLES AND RESPONSIBILITIES **6**

2.1. Headquarters Air Force. 6

2.2. Commanders/Directors of MAJCOM/FLDCOM, DRUs, and Field Operating Agencies. 13

Chapter 3—GOVERNANCE **23**

3.1. Overview. 23

3.2. Department of the Air Force Governance Process. 23

3.3. DAF Senior EMP Working Group (SEWG). 23

3.4. Reporting. 24

3.5. Mission Assurance Working Group (MAWG). 24

Chapter 4—REQUIREMENTS **25**

4.1. Overview. 25

4.2. Systems and Aircraft. 25

4.3. Facilities and Infrastructure. 27

4.4. Types of Tests. 28

Chapter 5—RISK DECISION **30**

5.1. Waiver Process (applicable to USAF only; USSF does not have a waiver process). 30

5.2. Waiver Cancellation Process. 30

Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION **31**

Chapter 1

INTRODUCTION

1.1. Background.

1.1.1. An electromagnetic pulse (EMP) is defined as a strong burst of electromagnetic radiation caused by a nuclear explosion, energy weapon, or by natural phenomenon, which may couple with electrical or electronic systems to produce damaging current and voltage surges. EMP events affect terrestrial, air, and space platforms and have_the capability to significantly disrupt, degrade, and damage electronic technology, critical measurement, and critical infrastructure systems. EMP includes those events particular to space-based systems, such as dispersed EMP (DEMP) and system generated EMP (SGEMP).

1.1.2. Human-made EMP events or naturally occurring (e.g., solar-induced space weather) Geomagnetic Disturbance (GMD) can affect large geographic areas, disrupting elements critical to the nation's security and economic prosperity, and could adversely affect global commerce and stability. The effects of an EMP range from temporary disruption of electronics technology and its functionality, to damaging specific electronic components, to disruption of the national electric power grid, to degradation of position, navigation, and timing. An EMP can significantly damage critical platforms and infrastructure, as well as the ability of the United States to project influence and military power. Protecting against these effects is a critical requirement each mission owner (as defined in the Terms) must consider and plan for as they evaluate their ability to perform required operations during and after EMP events.

1.2. Implementation.

1.2.1. This instruction provides the framework for EMP hardness assurance, incorporating EMP survivability testing and EMP hardness maintenance/hardness surveillance (HM/HS) programs to ensure systems and facilities are acquired, developed, designed, constructed, and sustained in accordance with applicable Military Standards (MIL-STD). The DAF will:

1.2.1.1. Design, test, and certify new mission-, safety-, and flight-critical systems, equipment, critical facilities, and infrastructure, using guidance provided in AFI 10-2607, DAFI 20-101_63-101, Department of Defense Instruction (DoDI) 5000.89, *Test and Evaluation*, and DAFI 99-103, *Capabilities-Based Test and Evaluation*, for EMP survivability during the acquisition or design/construction process.

1.2.1.2. Once the system is fielded and in the sustainment phase, periodically test the EMP survivability of the system and infrastructure, using guidance provided in AFI 10-2607 and AFPD 10-24.

1.2.1.3. Assess legacy and existing mission-, safety-, and flight-critical systems, equipment, critical facilities, and infrastructure; and ensure the most feasible approach for EMP survivability, including but not limited to retrofit, hardening system-required facilities, or rapid replacement of affected components and platforms.

1.2.1.4. Train personnel at the appropriate level to understand the unique requirements placed upon mission-critical systems and infrastructure within their organization.

1.2.1.5. Comply with EMP hardness technical requirements for both systems and infrastructure, as defined by MIL-STDs.

1.2.2. For purposes of this instruction, a mission-critical system is defined as a system whose operational effectiveness and operational capacity are essential to successful mission completion or to aggregate residual combat capability. If this system fails, the mission likely cannot be completed. Such a system can be an auxiliary or supporting system, as well as a primary mission system. Mission owners determine what systems are mission critical to support their mission needs. See AFI 10-2607, Figure 1, as required, to help inform mission critical determinations.

1.2.3. The ultimate goal of the EMP Survivability Program is to provide a realistic assessment of operational survivability and endurance through an EMP event. Examples of assessments are (not all inclusive) self-assessments, assessment and certification through the Integrated Chemical, Biological, Radiological, and Nuclear (CBRN) Survivability and Endurability Certification and Assessment Program for applicable units, and Balanced Survivability Assessments.

1.3. Objectives.

1.3.1. Identify roles and responsibilities for the DAF EMP Survivability Program to include:

1.3.1.1. Verification, as required for mission needs, and implementation of shortfall mitigation;

1.3.1.2. Establish a methodology for EMP test strategy and prioritization;

1.3.1.3. Establish EMP HM/HS program guidance for DAF capabilities to include systems and facilities; and

1.3.1.4. EMP survivability planning, programming, and budgeting.

1.3.2. Define a governance process to:

1.3.2.1. Report and characterize specific mission risks associated with EMP survivability testing and analysis non-compliance.

1.3.2.2. Identify EMP survivability shortfalls associated with all systems, infrastructure, and facilities requiring EMP survivability.

1.3.3. Identify education and training requirements for human-made EMP and naturally occurring GMD effects, subsequent recovery, vulnerabilities, and mitigation measures.

1.3.4. Establish/clarify EMP protection standards and critical prioritizations for mission design series aerospace vehicles and their associated support equipment, facilities, infrastructure, and other systems requiring EMP hardening, including nuclear command, control, and communications (NC3).

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. Headquarters Air Force.

2.1.1. **Assistant Secretary of Air Force for Acquisition, Technology, and Logistics (SAF/AQ).**

2.1.1.1. Ensure Program Executive Officers (PEO) comply with established EMP survivability requirements as identified by the Lead Command, as defined in the Terms section (e.g., JCIDS). Report EMP survivability shortfalls or risk acceptance to the Lead Command.

2.1.1.2. Ensure program managers develop, program for, and execute activities required to comply with EMP requirements.

2.1.1.3. Ensure program managers report requirement compliance status through the Lead Command to AFGSC, Air Force Materiel Command (AFMC), and Space Systems Command using procedures established by the Senior EMP Working Group (SEWG).

2.1.1.4. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.2. **Assistant Secretary of the Air Force for Space Acquisition and Integration (SAF/SQ)**

2.1.2.1. Ensure Program Executive Officers (PEO) comply with established EMP survivability requirements as identified by the Space Operations Command (SpOC) and Space Force Chief Operations Officer (SF COO/S10), as defined in the Terms section (e.g., Joint Capabilities Integration and Development System (JCIDS)). Report EMP survivability shortfalls or risk acceptance to the Lead Command.

2.1.2.2. Ensure program managers develop, program for, and execute activities required to comply with EMP requirements.

2.1.2.3. Ensure program managers report requirement compliance status through the service-designated EMP Lead Command to with procedures established by their respective service.

2.1.2.4. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.3. **Assistant Secretary of the Air Force for Financial Management and Comptroller (SAF/FM)**

2.1.3.1. Serve as the financial management representative to participate in the SEWG.

2.1.3.2. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.4. **Deputy Under Secretary of the Air Force for International Affairs (SAF/IA)**

2.1.4.1. Manage international interests regarding EMP requirements in Foreign Military Sales and other Security Cooperation activities.

2.1.4.2. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.5. **Assistant Secretary of the Air Force for Installations, Environment and Energy (SAF/IE)**

2.1.5.1. Serve as the Installation, Environment, and Energy representative for the EMP Survivability Program and serve as the OPR for programs and activities under their purview.

2.1.5.2. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.6. **Department of the Air Force Inspector General (SAF/IG)**

2.1.6.1. Ensure the Department of Air Force Inspection System supports United States Air Force (USAF) MAJCOM, and United States USSF FLDCOM independent, risk-based inspections and assessments of critical combat missions and weapon systems linked to the EMP Survivability Program.

2.1.6.2. Ensure the Inspection System's Headquarters Inspection Program assesses the overall effectiveness and efficiency of the EMP Survivability Program.

2.1.6.3. Advise the SEWG on DAF inspection policy using guidance provided in DAFI 90-302, *The Inspection System of the Department of the Air Force*, and, as necessary, provide access to inspection feedback on unit EMP survivability, unit capability assessments, and overall effectiveness of the DAF EMP Survivability Program.

2.1.6.4. The Inspector General, Office of Special Investigations, will provide proactive and comprehensive counterintelligence support.

2.1.7. **Director for Studies and Analyses (SAF/SA)**

2.1.7.1. Provide analytic insights to support the countering EMP Survivability program through the application of analytical techniques such as mission-level and campaign-level modeling analysis.

2.1.7.2. In coordination with Deputy Chief of Staff for Air Force Futures (AF/A5/7), A10, SF COO/S10, Deputy Chief of Space Operations for Strategy, Plans, Programs and Requirements (CSRO), and USSF Space Warfighting Analysis Center (SWAC) integrate relevant USAF and USSF input for countering EMP into Support for Strategic Analysis products (e.g., scenarios, concepts of operation, and multi-service force deployments), using guidance in DoDD 8260.05, *Support for Strategic Analysis (SSA).*

2.1.8. **Deputy Chief of Staff for Intelligence (AF/A2)**

2.1.8.1. Provide policy, guidance, and oversight for operations, training, and sourcing of USAF intelligence, surveillance, reconnaissance, and personnel to counter EMP operations.

2.1.8.2. Provide comprehensive and timely reporting of potential foreign threat incidents, events and trends to the SEWG.

2.1.8.3. Plan, program, and budget for resources to support DAF countering EMP activities.

2.1.8.4. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.9. **Deputy Chief of Staff for Operations (AF/A3)**

2.1.9.1. Manage and maintain overall responsibility for implementing Mission Assurance and the Mission Assurance Construct, including managing risk to the USAF Defense Critical Infrastructure list, using guidance provided in AFPD 10-24 and DoDI 3020.45-V1, *Mission Assurance Construct*. This includes treating EMP and its effects as part of CBRN mission sets.

2.1.9.2. Advocate for continuity of operations (COOP) program planning using guidance provided in DAFI 10-208, *Continuity of Operations (COOP) Program*.

2.1.9.3. Ensure Mission Essential Tasks (METs) are developed to facilitate deficiency reporting in Defense Readiness Reporting System, using guidance provided in AFI 10-201, *Force Readiness Reporting* and SpFI 10-201, *Force Readiness Reporting*.

2.1.9.4. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.9.5. Provide space weather monitoring, analysis, and prediction operations and capabilities in support of all elements of the DoD and DoD Intelligence Components, using guidance provided in Chairman of the Joint Chiefs of Staff Instructions (CJCSI) 3810.01G, *Meteorological and Oceanographic Support,* Enclosure B-5, paragraph 5.b.

2.1.10. Deputy Chief of Staff for Logistics, Engineering, and Force Protection (AF/A4)

2.1.10.1. Integrate established EMP HM/HS requirements for logistics to include support equipment, mission critical facilities and infrastructure in the appropriate maintenance and logistics guidance.

2.1.10.2. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.11. Deputy Chief of Staff for Air Force Futures (AF/A5/7)

2.1.11.1. Advise the SEWG on strategic requirements and provide recommendations on the integration of counter EMP operational requirements using guidance provided in AFI 10-601, *Operational Capability Requirements Documentation and Validation*.

2.1.11.2. Represent the USAF to the joint community on counter EMP operational requirements through the JCIDS and the Joint Requirements Oversight Council (JROC) using guidance provided in AFI 10-601.

2.1.11.3. Integrate considerations for EMP effects into support for Strategic Analysis products (e.g., scenarios, concepts of operation, and multi-service force deployments), using guidance provided in DoDD 8260.05.

2.1.11.4. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.12. **Deputy Chief of Staff for Warfighter Communications & Cyber Systems (AF/A6)**

2.1.12.1. Provide policy, guidance, and oversight for operations, training, and sourcing of USAF communications and cyber effects capabilities and personnel to counter EMP operations.

2.1.12.2. Provide comprehensive and timely reporting of potential cyber threat incidents, events and trends to the SEWG.

2.1.12.3. Plan, program, and budget for resources to support DAF countering EMP activities.

2.1.12.4. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.13. **Deputy Chief of Staff for Plans and Programs (AF/A8)**

2.1.13.1. Support USAF planning, programming and allocation of associated resources for the EMP Survivability Program.

2.1.13.2. Advise the SEWG in the programming, planning, budgeting, and execution process for EMP related efforts.

2.1.13.3. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.14. **Deputy Chief of Staff for Strategic Deterrence and Nuclear Integration (AF/A10)**

2.1.14.1. Serve as the Headquarters Air Force Accountable Officer responsible for the USAF Nuclear Enterprise and provide direction, guidance, integration, and advocacy regarding the nuclear deterrence mission of the USAF. This includes serving as final approval authority for USAF EMP Test Strategy.

2.1.14.2. Ensure significant EMP issues are brought to the attention of the SecAF, Chief of Staff of the Air Force (CSAF) and Chief of Space Operations (CSO), and coordinate resolution through appropriate governance bodies, such as the Nuclear Oversight Council. Additionally, the biannual Nuclear Mission Assessment is another avenue for reporting significant EMP issues to DAF leadership.

2.1.14.3. Support MAJCOM/FLDCOM submission for required resources to execute the DAF EMP Test Strategy.

2.1.14.4. Serve as SEWG tri-chair with AFGSC A3/6 and SF COO/S10.

2.1.14.5. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.14.6. Ensure survivability requirements are incorporated into applicable requirements, resourcing, and acquisition policy through coordination with DAF Requirements and Acquisition Communities.

2.1.14.7. Provide DAF Corporate Structure information to support adequate resourcing of EMP survivability requirements.

2.1.14.8. Be responsible for establishing and publishing all DAF EMP policies to include interface with all CBRN Survivability requirements.

2.1.14.9. Be responsible for facilitating EMP Test Strategy execution across the DAF.

2.1.14.10. Facilitate and integrate EMP Survivability Program issues through Nuclear Oversight Council or to other governance bodies, as required.

2.1.14.11. Facilitate and address EMP Survivability Program issues across the Services, Joint community, and Interagency partners.

2.1.14.12. Develop an EMP prioritization model of critical facilities and infrastructure in coordination with AF/A3, AF/A4, AF/A5, SAF/SA, Deputy Chief of Space Operations for Operations, Cyber and Nuclear (SF COO), SF/S5/8, SAF/AQ, SAF/SQ, and US Strategic Command (USSTRATCOM), along with respective mission owners and Air Force Installation and Mission Support Center (AFIMSC). This model will be based on continuity of operations or revised existing models for all critical facilities and infrastructure, either in whole or parts critical to the mission success as a part of their mission assurance. Each mission owner's prioritization list will be included as part of the EMP annual report.

2.1.14.13. Lead coordination with other mission areas (AF/A2, AF/A3, AF/A4, AF/A6, etc.) to implement the continuity of operations.

2.1.14.14. Advocate for real property and non-RPIE requirements for NC3 and Command, Control, Communications, Computers, and Intelligence (C4I) facilities to mitigate against EMP effects, sustainment, and maintenance shortfalls.

2.1.14.15. In coordination with AF/A3, ensure deficiencies affecting the EMP hardness of required systems are reported in Defense Readiness Reporting System.

2.1.15. **The Office of the Judge Advocate General (AF/JA).** Advise, through the Operations and International Law Directorate, the EMP Enterprise in the development and review of strategy, plans, policies, procedures, training and operations relating to countering EMP, including counter-proliferation interdiction and Proliferation Security Initiative activities, to ensure consistency with U.S. domestic law and the obligations of the United States under international law, including the Law of War and other relevant treaties, international agreements and host-nation laws.

2.1.16. **The Chief of Air Force Reserve (AF/RE)**

2.1.16.1. Be the lead for EMP Reserve actions and activities as required.

2.1.16.2. Identify and develop, based on each critical mission, a list of all Air Force Reserve equipment, facilities, and infrastructure critical to the Air Force Reserve Command program that would be affected by an EMP event.

2.1.16.3. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.17. **Air Force Surgeon General (AF/SG).** Provide policy, guidance, and oversight for operations, training, readiness, and sourcing of DAF medical capabilities and personnel to support countering EMP operations.

2.1.18. **Director of Test and Evaluation (AF/TE).**

2.1.18.1. Establish test and evaluation policy, advocate for test and evaluation resources required to support EMP survivability development, identify sustainment costs, and resolving test and evaluation issues and disputes.

2.1.18.2. Collaborate with requirements sponsors and system developers to improve the development, testing, and fielding of DAF systems or subsystems subjected to EMP survivability testing.

2.1.18.3. Fulfill any other responsibilities covered in DoDI 5000.89_DAFI 99-103 applicable to EMP survivability testing.

2.1.18.4. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.19. **Deputy Chief of Space Operations for Intelligence (SF/S2)**

2.1.19.1. Provide policy, guidance, and oversight for operations, training, and sourcing of USSF intelligence, surveillance, and reconnaissance (ISR) capabilities and Integrated Tactical Warning and Attack Assessment (ITW/AA) weapon system including personnel to enable countering EMP operations

2.1.19.2. Provide comprehensive and timely reporting of potential foreign threat incidents, events and trends to the SEWG.

2.1.19.3. Plan, program, and budget for resources to support DAF countering EMP activities.

2.1.19.4. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.20. **Deputy Chief of Space Operations for Operations, Cyber, and Nuclear (SF/3/4/7/10 as SF COO/S10)**

2.1.20.1. Be the lead for the EMP Survivability program by defining, establishing, and executing the program.

2.1.20.2. Provide policy, guidance, and oversight for operations, training, and sourcing of USSF NC3 capabilities and Integrated Tactical Warning and Attack assessment systems (ITW/AA) including personnel to enable countering EMP operations.

2.1.20.3. Identify current EMP detection and characterization capabilities and ensure future space-based EMP detection systems fulfill requirements outlined in this DAFI.

2.1.20.4. Through the SF COO/S4: integrate established EMP survivability testing requirements for logistics (to include support equipment) mission critical facilities and infrastructure in the appropriate maintenance and logistics guidance.

2.1.20.5. Through SF COO/S10:

2.1.20.5.1. Provide direction, guidance, integration, and advocacy for the USSF provisioned NC3 Deterrence mission/Nuclear Enterprise.

2.1.20.5.2. Establish the USSF EMP Program. S10 will develop and independently manage the USSF EMP program (independence implementation window not to exceed 5 years). Initially, upon the completion of the standup, the USSF EMP Program will be responsible for advocacy, oversight of mission requirements prioritization, policy and guidance, and resourcing processes and procedures. Final scope of the program oversight will be developed with support from SpOC and subordinate units where applicable.

2.1.20.5.3. Ensure significant EMP issues are brought to the attention of the SecAF and Chief of Space Operations (CSO) and coordinate resolution through appropriate governance bodies, such as the Nuclear Oversight Council. Additionally, the biannual Nuclear Mission Assessment is another avenue for reporting significant EMP issues to DAF leadership.

2.1.20.5.4. Support FLDCOM submissions for required resources to execute the DAF EMP Test Strategy.

2.1.20.5.5. Serve as SEWG tri-chair with AFGSC A3/6, AF/A10C until USSF completes standing up USSF EMP Program.

2.1.20.5.6. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.20.5.7. Ensure survivability requirements are incorporated into applicable requirements, resourcing, and acquisition policy through coordination with DAF Requirements and Acquisition Communities.

2.1.20.5.8. Provide USSF Corporate Structure information to support adequate resourcing of EMP survivability requirements.

2.1.20.5.9. Be the USSF OPR responsible for establishing and publishing all DAF EMP policies to include interface with all CBRN Survivability requirements.

2.1.20.5.10. Be responsible for facilitating EMP Test Strategy execution across the USSF.

2.1.20.5.11. Facilitate and integrate EMP Survivability Program issues through Nuclear Oversight Council or to other governance bodies, as required.

2.1.20.5.12. Address USSF EMP Survivability Program issues across the Services, Joint community, and Interagency partners.

2.1.20.5.13. Develop an EMP prioritization model of critical facilities and infrastructure in coordination with the multiple roles of the Deputy Chief of Space Operations for Operations, Cyber, and Nuclear SF/S5/8, SAF/SA, SAF/SQ, and US Strategic Command (USSTRATCOM), along with respective mission owners and USSF Installation Commanders. This model will be based on COOP, the mission assurance construct, or revised existing models for all critical facilities and infrastructure, either in whole or parts critical to the mission success as a part of their mission assurance. Each mission owner's prioritization list will be included as part of the EMP annual report.

2.1.20.5.14. Advocate for real property and non-RPIE requirements for NC3 and C4I facilities to mitigate against EMP effects, sustainment, and maintenance shortfalls.

2.1.21. **Deputy Chief of Space Operations for Cyber and Data (SF S6)**

2.1.21.1. Advise the SEWG and provide recommendations on the integration of EMP survivability requirements in development of future USSF capabilities.

2.1.21.2. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.1.22. **Deputy Chief of Space Operations for Strategy, Plans, Programs, and Requirements (SF S5/8)**

2.1.22.1. Advise the SEWG on strategic requirements and provide recommendations on the integration of countering EMP operational requirements.

2.1.22.2. Represent the USSF to the joint community on countering EMP operational requirements through the JCIDS and the JROC using guidance provided in AFI 10-601.

2.1.22.3. Integrate considerations for EMP effects into support for Strategic Analysis products (e.g., scenarios, concepts of operation, and multi-service force deployments), in accordance with DoDD 8260.05.

2.1.22.4. Support USSF planning, programming and allocation of associated resources for the EMP Survivability Program.

2.1.22.5. Advise the SEWG in the programming, planning, budgeting, and execution process for EMP related efforts.

2.1.23. **Director, Air National Guard (NGB/CF)**

2.1.23.1. Be the lead for EMP Air National Guard actions and activities as required.

2.1.23.2. Identify and develop, based on each critical mission, a list of all Air National Guard-related equipment, facilities, and infrastructure critical to the Air National Guard program that would be affected by an EMP event.

2.1.23.3. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG.

2.2. Commanders/Directors of MAJCOM/FLDCOM, DRUs, and Field Operating Agencies.

2.2.1. USAF units will monitor EMP survivability compliance of assigned systems and owned facilities through assessments, Program Management Officers (PMOs), AFIMSC, Program Executive Offices, and Air Force Nuclear Weapon Center (AFNWC), as required by AF/A10. USSF units will monitor EMP survivability compliance of assigned systems and owned facilities, as required by SF COO.

2.2.2. Assess mission impact of reported EMP survivability shortfalls and/or noncompliance with tests and analysis, resource or assume risk in a prioritized and informed manner, and inform AFGSC/AFNWC (for USAF units) and SpOC (for USSF units) of risk decisions made, as required by AF/A10 or SF COO/S10, respectively.

2.2.3. Provide AFGSC and AFNWC a prioritized list of threat-level testing of mission-critical equipment and aircraft, as required by AFGSC.

2.2.4. Appoint an OPR to address EMP-related issues and participate in the DAF EMP governance process through the SEWG. When USSF completes standing up its own EMP governance process, the OPR will join USAF EMP governance process.

2.2.5. Program for EMP. Coordinate with AFGSC to fund survivability assessments.

2.2.6. Ensure EMP-related threats and hazards are integrated into installation All Hazard Threat Assessment and provide risk response mitigation measures to vulnerabilities found during assessments.

2.2.7. Ensure EMP survivable equities are included in the planning phase of exercises and assessments.

2.2.8. **Commanders of Lead Commands** ("Lead Command" as defined in DAFPD 10-9, *Lead Command/Lead Agent Designation and responsibilities for United States Air Force Weapon Systems, Non-Weapon Systems, and Activities*)

2.2.8.1. Appoint an OPR to address EMP-related issues.

2.2.8.2. Provide representation to each level of the governance process.

2.2.8.3. Identify critical systems and support equipment that must survive an EMP event.

2.2.8.4. Provide a mission-based prioritized list of assigned weapons system to AFGSC and AFNWC. The list will include whether EMP hardness of each asset is known or unknown. This list will be reassessed on a recurring basis, not less than every 2 years.

2.2.8.5. Program for less-than-threat-level EMP assessments, HM/HS, and mitigations, as applicable.

2.2.9. **Commander, Air Force Global Strike Command (AFGSC/CC)**

2.2.9.1. Serve as the USAF EMP Survivability Testing Oversight Coordinator.

2.2.9.2. Advocate for resources to conduct modeling, analysis, and EMP survivability testing in support of the Lead Command.

2.2.9.3. Track and provide oversight of lower than threat system-level (e.g., continuous wave immersion for aircraft) testing in support of supporting the asset-owner Lead Command.

2.2.9.4. Submit Planning and Programming Options on funding (to include Manpower billets, if applicable) to the Air Force Corporate Structure for all Air Force threat-level EMP survivability testing for existing systems, and new systems, once fielded, using guidance provided in the DAF EMP Test Strategy.

2.2.9.5. Establish, and along with AF/A10 and SF COO/S10, tri-chair the DAF SEWG through AFGSC/A3/6. Through the SEWG:

2.2.9.5.1. Collect annual reports from each MAJCOM/FLDCOM on the status of compliance of critical systems and facilities to EMP requirements, risk-acceptance decisions and associated mitigation actions, EMP survivability testing activities, and resourcing concerns.

2.2.9.5.2. Provide to AF/A10, SF/COO, MAJCOMs, FLDCOMs, AFNWC, and other organizations, the recommended priority lists on EMP survivability testing and potential mitigation actions, to inform and guide potential resourcing actions. SecAF is the final approval authority of the priority lists.

2.2.9.5.3. The AFGSC/A3/6 provides the direction of USAF EMP survivability testing of existing systems and risk-acceptance decisions of AFGSC assets.

2.2.9.5.4. Annual USAF-level EMP reports are the responsibility of the AFGSC in support of the Nuclear Oversight Council, or other governance bodies, as required.

2.2.9.5.5. Consolidate mission-based prioritization lists from commanders of MAJCOMs, FLDCOMs, DRUs, and Field Operating Agencies. AFGSC/CC will then provide a consolidated list to SecAF via Nuclear Oversight Council, Nuclear Mission Assessment, or other appropriate means.

2.2.10. **Commander, Air Force Materiel Command (AFMC)**

2.2.10.1. Provide a senior leader (O-6/GS-15) and action officer to participate in the SEWG. AFIMSC, AFNWC, Air Force Research Laboratory, and Air Force Life Cycle Management Center, and other AFMC Centers/Agencies will serve as advisors to Headquarters (HQ) AFMC, but do not have voting privileges in the SEWG.

2.2.10.2. Identify resource shortfalls for threat-level EMP test requirements annually to AFGSC. Report program-level EMP requirements and capabilities to AFGSC to inform EMP requirements, development, verification, and sustainment.

2.2.10.3. **Through AFNWC**

2.2.10.3.1. Develop the DAF-wide EMP Test Strategy through coordination with USSF, AFGSC, Lead Commands, Program Executive Offices, and program offices, as applicable.

2.2.10.3.2. Once capability exists, execute the DAF EMP Test Strategy including organic USAF test capability development, asset and test facility availability coordination, data review and analysis, problem identification, and risk analysis, and participates in mitigation strategy development.

2.2.10.3.3. Partner with mission owners for EMP survivability sustainment and testing of critical facilities and infrastructure to identify what assets require testing and to ensure EMP survivability is characterized. Additionally:

2.2.10.3.3.1. Ensure new or emerging EMP survivability is characterized on existing critical systems.

2.2.10.3.3.2. Document test results in appropriate mission assurance (MA) repositories.

2.2.10.3.3.3. Identify resource requirements, shortfalls, and associated prioritization to the owning MAJCOM/FLDCOM for budget, programming, or resource action.

2.2.10.4. **Through AFIMSC**

2.2.10.4.1. Upon request, provide real property facility expertise to the SEWG while supporting EMP survivability requirements to assist MAJCOMs and FLDCOMs in priority assessment of shortfall mitigation.

2.2.10.4.2. Partner with mission owners and stakeholders to provide a standardized approach to EMP survivability and testing compliance using guidance provided in MIL-STDs for all applicable DAF facilities requiring protection.

2.2.10.4.3. Support the identification of resource shortfalls and associated prioritization to the owning MAJCOM/FLDCOM for resource requirements exceeding programmatic funding.

2.2.10.4.4. Provide a standardized approach to EMP survivability and testing compliance using guidance provided in MIL-STDs for all applicable USAF facilities requiring protection.

2.2.10.4.5. Review and coordinate on EMP hardening and protective measures for all infrastructure and facility infrastructure assets as determined by mission-based priority lists.

2.2.10.4.6. Advocate for real property and non-RPIE requirements for NC3 facilities to mitigate against EMP protection, sustainment, and maintenance shortfalls.

2.2.10.4.7. Implement an EMP prioritization model developed by AF/A10 and SF COO/S10, based on Continuity of Operations, in coordination with AF/A10, AF/A5, SF COO/S10, SF/S5 and respective mission owners. AFIMSC will incorporate the models critical to the mission success for all facility and infrastructure either in whole or part as a part of their mission assurance.

2.2.10.5. **Through Air Force Life Cycle Management Center and Air Force Program Offices**

2.2.10.5.1. Resource Program Offices to execute the program's EMP Test Strategy, including asset and test facility availability coordination, data review and analysis, problem identification, and risk analysis, and participates in mitigation strategy development.

2.2.10.5.2. Provide weapons system expertise to the EMP Working Group on system and facility EMP survivability requirements to assist MAJCOMs and FLDCOMs in priority assessment of shortfall mitigation.

2.2.10.5.3. Partner with mission owners for EMP survivability sustainment and testing of critical systems to assess the results of testing and to ensure EMP survivability requirements are met. Identify resource shortfalls and associated prioritization to the owning MAJCOM/FLDCOM or AFGSC/SpOC for resource requirements exceeding programmatic funding.

2.2.10.5.4. Coordinate with Lead Commands as applicable to fuse operational mission impact and warfighting characterizations resulting from EMP test and assessment results on existing systems to create a priority list of required mitigation or corrective actions. Provide inputs to the SEWG's annual plan to AF/A10, SF/COO, MAJCOMs/FLDCOMs for potential resourcing action.

2.2.10.5.5. Identify resource shortfalls for threat-level EMP test requirements annually to AFGSC and SpOC.

2.2.10.5.6. Support System Program Office requirement to develop EMP test plans and implement weapon system testing as part of acquisition and sustainment.

2.2.11. **Commander, Air Education and Training Command (AETC/CC) will:**

2.2.11.1. Ensure a continuum of learning regarding EMP survivability requirements, methods, testing and evaluation, research, and acquisition throughout all levels of nuclear training and Professional Military Education (PME). EMP training and education should be multidisciplinary and sustained in order to establish a strong culture of EMP maintenance procedure compliance. Exception to this are engineers within the program offices since their training is highly technical and will be primarily on-the-job training.

2.2.11.2. Integrate education on EMP survivability and maintenance measures into applicable nuclear training courses, Electromagnetic Spectrum Operations training, initial skills courses, and PME, as required to create a force knowledgeable of EMP effects on mission readiness and continuation of operations following a human-made or naturally occurring EMP event.

2.2.11.3. Establish on-time, on-demand training and education resources to raise awareness of EMP effects and mitigation techniques and maintain skills and knowledge necessary to survive an EMP event.

2.2.11.4. Develop and advocate for incorporation of EMP events into USAF war games, exercises, and training scenarios.

2.2.12. **Commander, Space Operations Command (SpOC/CC)**

2.2.12.1. Maintain oversight of all Integrated Tactical Warning/Attack Assessment (ITW/AA) and NC3 assets within the USSF portfolio.

2.2.12.2. Submit Planning and Programming Options on funding (to include Manpower billets, if applicable) to the USSF Corporate Structure for all USSF threat-level EMP survivability testing for existing systems, and new systems, once fielded, using guidance provided in the USSF EMP Test Strategy.

2.2.12.3. Provide oversight and management of the USSF Survivable/Endurable (S/E) Program. The S/E Program provides management, guidance, policy, assessment, operational prioritization , advocacy, and instruction for survivable/endurable USSF equities to include NC3-ITW/AA, and ISR assets within its portfolio. Program oversight will be developed with support from AFGSC, AF/A10, SF COO/S10, and subordinate centers where applicable. Through the S/E Program:

2.2.12.3.1. Maintain lead command oversight of all USSF EMP hardened missions to include NC3, ITW/AA, and ISR assets.

2.2.12.3.2. Identify critical systems and support equipment that must survive an EMP event.

2.2.12.3.3. Collect from each Mission Delta (MD) quarterly reports on the status of EMP protected critical systems and facilities, risk-acceptance decisions and associated mitigation actions, EMP survivability testing activities, and resourcing concerns.

2.2.12.3.4. Provide a prioritized list of critical systems and infrastructure supporting USSF EMP protected missions to SF COO/S10.

2.2.12.3.5. Provide SF COO/S10 with a prioritized list of required mitigation or corrective actions through coordination with FLDCOMs, Deltas, and units as applicable to fuse operational mission impact and warfighting characterizations stemming from EMP test and assessment results on existing USSF systems.

2.2.12.3.6. Provide SF COO/S10 with an annual report on USSF equities derived from USSTRATCOM's Integrated Nuclear Survivable/Endurable Report (INSER) and Integrated CBRN Survivability and Endurability Certification and Assessment Program (ICSECAP) assessment reports, Defense Threat Reduction Agency (DTRA) Mission Assurance Assessment (MAA) reports, and USSF Inspector General assessment reports containing survivable/endurable elements.

2.2.12.3.7. Provide Oversight and Management of the USSF EMP Test Strategy as prioritized and directed by SF COO/S10.

2.2.12.3.8. Assess and report mission impact of USSF EMP survivability shortfalls and/or non-compliant EMP test results. Advocate to resource or assume risk in a prioritized and informed manner.

2.2.12.3.9. Coordinate with FLDCOM/Delta/Unit S2 offices to integrate EMP-related threats and hazards are into installation all hazard threat assessment and provide risk response mitigation measures to vulnerabilities found during assessments.

2.2.12.3.10. Advocate for resources to conduct USSF EMP modeling, analysis, and survivability testing in support of SF COO/S10.

2.2.12.3.11. Advocate for EMP events into USSF war games, exercises, and training scenarios.

2.2.12.3.12. Ensure EMP survivable equities are included in the planning phase of exercises and scheduled mission assurance related assessments.

2.2.12.3.13. Report on SpOC EMP survivability in the Defense Readiness Reporting System to include plans, programs, and procedures developed to mitigate EMP survivability shortfalls. SpOC/S10S will also coordinate with SpOC/S735R to ensure published updates to national, DoD, joint staff, CCMD, and DAF level EMP survivability policies are accurately captured in USSF Survivable/Endurable MET measures.

2.2.12.3.14. Support USSF Mission Delta and units with EMP survivability inputs into SpOC corporate process for resourcing (i.e. Program of Memorandum (POM)); Support includes providing expertise, guidance, and advocacy in the financial corporate process for applicable organizations (SpOC, AFSVC, AFIMSC, etc.) and throughout the USSF corporate process in coordination with SF COO/S10.

2.2.12.3.15. In support of the HEMP PMO within Space System Command (SSC), advocate for SpOC funding of EMP protection as an operational capability for SpOC Mission Deltas and units IAW national policies.

2.2.12.3.16. Serve as executive agent for SF COO/S10 in the USSF SEWG. Through the USSF SEWG:

2.2.12.3.16.1. Provide to SF COO/S10, MAJCOMs, SSC, and other organizations, the adjudicated annual priority lists on EMP survivability testing and mitigation actions, to inform and guide potential resourcing actions. SpOC/S10S leads SpOC/CC's analyses, assessments, and special studies to support adjudication recommendations on mitigations, which may involve protection of equipment and/or facilities, or a combination of the two.

2.2.12.3.16.2. SpOC/S10S provides the direction of USSF EMP survivability testing of existing systems and risk-acceptance decisions of SpOC assets made by SpOC/CC.

2.2.12.3.16.3. Annual USSF-level EMP reports are the responsibility of SpOC/S10S in support of the Nuclear Oversight Council, or other governance bodies, as required.

2.2.12.3.16.4. Consolidate mission-based prioritization lists from Commanders of FLDCOMs, DRUs, and Field Operating Agencies. SpOC/S10S will provide to SpOC/CC a report of a holistic view of DAF health in terms of EMP hardness/protection status based on the received consolidated list. The consolidated list will also be submitted to the Nuclear Oversight Council as required.

2.2.12.3.16.5. Report on EMP survivability compliance of assigned systems and owned facilities from data provided by the Mission Assurance Assessment program, Integrated CBRN Survivability and Endurability Certification and Assessment Program (ICSECAP), Program Management Offices (PMOs), Program Executive Offices, as required by SF COO/S10.

2.2.12.3.16.6. Provide a mission impact analysis of reported EMP survivability shortfalls and/or noncompliance with tests and analysis, and advocate for SF COO/S10 decision to resource or assume risk in a prioritized and informed manner.

2.2.12.4. Maintain and provide upon request, a repository of current national, DOD, CJCS, DAF and CCMD policies, ICDs/CCDs, system architecture documents, MIL-STDs, and training materials to educate USSF operations personnel on the impacts of EMP effects on mission readiness and continuation operations following an EMP event.

2.2.12.5. Establish on-time, on-demand training and education resources to raise awareness of EMP effects, operational impacts, USSF S/E Construct, mitigation techniques, and COOP/COG to maintain skills and knowledge necessary to survive an EMP event.

2.2.13. **Commander, Space Systems Command (SSC/CC)**

2.2.13.1. Manage High Altitude EMP (HEMP) HEMP sustainment of all USSF EMP protected systems to include NC3, and ITW/AA assets within the USSF portfolio.

2.2.13.2. Provide oversight and management of the USSF HEMP Program Management Office (HPMO). The HPMO plans and programs EMP hardening, refreshment/upgrades, new HEMP protected builds, HM/HS, and verification testing of USSF EMP protected equities. The HPMO serves as the configuration manager for HEMP topologies under its purview. The final scope of the program oversight will be developed with support from SF COO/S10, SpOC, and subordinate centers where applicable. Through the HPMO:

2.2.13.2.1. Construct, implement, and manage contracts for sustainment, testing, and upgrades.

2.2.13.2.2. Develop a standardized approach for creating and implementing HM/HS plans for each USSF EMP protected asset.

2.2.13.2.3. Work with SpOC/S10, SSC PMOs, SpRCO, SDA, Space Futures Command (when activated), AFRL (space programs), and non-USSF entities with space equities (e.g., AF RCO) to advocate for funding to SpOC, SF COO/S10, and SF S/5/8.

2.2.13.2.4. Coordinate with HEMP managers at USSF EMP protected sites to ensure HM/HS is conducted with minimal impact on the mission.

2.2.13.2.5. Serve as configuration management for HEMP topologies of USSF EMP protected equities.

2.2.13.2.6. Provide a quarterly report on the status of HEMP construction, sustainment, testing, and configuration management activities to the SpOC S/E Program and SF COO/S10 for SEWG reporting, integration into operational planning, and funding advocacy.

2.2.13.2.7. Implement the USSF EMP Test Strategy for USSF EMP validation through coordination with SF COO/S10, SpOC/S10, and all USSF Program Executive Offices and program offices, as applicable.

2.2.13.2.8. Provide, upon request, the relevant expertise on system and facility EMP survivability requirements to FLDCOMs, (including Space Futures Command, when activated), Space RCO, SDA, Lead Commands, AFRL (space programs), and non-USSF entities with space equities (e.g., AF RCO) to assist in programming, planning and implementation of EMP protected facility projects/contracts/upgrades.

2.2.13.2.9. Synchronize MIL-STD requirements across the USSF portfolio through active involvement with USSF projects to construct and remediate EMP protected facilities.

2.2.13.2.10. Conduct HEMP survivability studies as required to inform HEMP projects, upgrades, remediation, and configuration changes

2.2.13.2.11. Train HEMP Managers at direct report units (DRUs) on HM/HS, HM/HS contracts, and HEMP requirements.

2.2.13.2.12. Implement an EMP prioritization model developed by SF COO/S10, based on Continuity of Operations, in coordination with SF COO/S10, SF/S5 , SpOC/S10S, and respective mission owners. AFIMSC DET 1 will incorporate the models critical to the mission success for all facility and infrastructure either in whole or part as a part of their mission assurance.

2.2.13.3. Maintain a repository of current MIL-STDs, Tech performance criteria, CCMD and DAF policies pertaining to the design and implementation of HEMP protection to educate contract officers, sustainment personnel and HEMP managers on the impacts of EMP effects on mission readiness and protection/mitigation techniques used to prevent mission impact.

2.2.13.4. Train HEMP Managers at direct report units (DRUs) on HM/HS, HEMP contracts (SOW, QA, roles and responsibilities), and HEMP sustainment/mitigation techniques.

2.2.14. **Commander, Space Training and Readiness Command (STARCOM/CC)**

2.2.14.1. Develop and advocate for EMP events into USSF war games, exercises, and training scenarios.

2.2.14.2. STARCOM does not train EMP maintenance or provide any weapon system specific EMP survivability training.

2.2.15. **USSF Installation/Delta/Unit Commanders who oversee assets, sites or platforms with EMP survivability requirements will:**

2.2.15.1. Appoint a HEMP Manager to oversee the HEMP topology and coordinate unit level HM/HS activities. The HEMP Manager will:

2.2.15.1.1. Coordinate between the unit and SSC HPMO to enable timely execution of an active and effective HM/HS Program with minimal impact to the mission.

2.2.15.1.2. Ensure respective EMP protective sub-systems are maintained throughout their life cycle (equipment, facilities, platforms).

2.2.15.1.3. Identify and report deficiencies and HEMP topology configuration changes to the SSC HPMO.

2.2.15.1.4. Ensure site personnel receive EMP survivability education and training.

2.2.15.1.5. Ensure site-specific plans and TTPs are developed to effectively assure survivable, recoverable, and endurable mission functions.

2.2.15.1.6. Support Emergency Management in the development of comprehensive logistics support plan to enable each respective mission to meet survivability and endurability requirements.

2.2.15.1.7. Report proposed configuration changes affecting HEMP topology to HPMO within SSC.

2.2.15.2. Ensure the HEMP Manager is included in all unit level discussions that potentially change the configuration of EMP protected ground facilities/mobiles.

2.2.15.3. Maintain a repository of MIL-STDs, Technical Orders (TOs), and training pertaining to the care and operator maintenance of HEMP protected features of the site.

2.2.15.4. Develop basic EMP survivability education and training (HEMP 101) on local HEMP topology operation and care and ensure unit personnel are trained upon assignment and annually to cultivate a USSF culture of HEMP readiness minded guardians.

2.2.16. **USAF Installation/Wing Commanders who oversee assets, sites or platforms with EMP survivability requirements**

2.2.16.1. Appoint a Hardness Maintenance Hardness Surveillance (HM/HS) Program Manager (PM).

2.2.16.1.1. Oversee execution of an active and effective HM/HS program. The HM/HS PM will:

2.2.16.1.2. Develop a repository to record the following actions:

2.2.16.1.2.1. Ensure respective EMP protective sub-systems are maintained throughout their life cycle (equipment, facilities, platforms).

2.2.16.1.2.2. Identify and report deficiencies to MAJCOM representatives.

2.2.16.1.2.3. Ensure site personnel receive EMP and CBRN survivability education and training.

2.2.16.1.2.4. Ensure site-specific plans and TTPs are developed to effectively assure survivable, recoverable, and endurable mission functions.

2.2.16.1.2.5. In coordination with base emergency offices, support the development of comprehensive logistics support plan to enable each respective mission to meet survivability and endurability requirements.

2.2.16.2. Develop the USAF EMP Test Strategy through coordination with USAF, MAJCOMs, Program Executive Offices, and program offices, as applicable.

2.2.16.3. Once capability exists, execute the USAF EMP Test Strategy including organic USAF test capability development, asset and test facility availability coordination, data review and analysis, problem identification, and risk analysis, and participates in mitigation strategy development.

2.2.16.4. Partner with mission owners for EMP survivability sustainment and testing of critical facilities and infrastructure to identify what assets require testing and to ensure EMP survivability is characterized. Additionally:

2.2.16.5. Ensure new or emerging EMP survivability is characterized on existing critical systems.

2.2.16.6. Document test results in appropriate mission assurance (MA) repositories.

2.2.16.7. Identify resource requirements, shortfalls, and associated prioritization to the owning MAJCOM for budget, programming, or resource action.

Chapter 3

GOVERNANCE

3.1. Overview. The DAF EMP Survivability Program governance process is designed to inform MAJCOM/FLDCOM and DAF senior leaders of the health and status of EMP survivability and quantify risk to DAF systems, equipment, facilities, and infrastructure. Transparency of survivability status and risk acceptance at the appropriate level is a vital component to the governance process.

3.2. Department of the Air Force Governance Process. Unless otherwise outlined in this DAFI, the approval process will adhere to the DAF Governance Process.

3.2.1. All issues requiring higher level authorities will be brought before the SEWG.

3.2.2. The independent service SEWG's will determine if the issue should be brought before AFGSC/CC and AF/A10 Director for USAF equities, and SpoC/CC and SF COO/S10 Director for USSF equities, who then has authority to elevate to Executive Leadership (SecAF, CSAF, etc.) through the Nuclear Oversight Council.

3.3. DAF Senior EMP Working Group (SEWG). The purpose of the SEWG is to be the DAF-wide governing body on EMP survivability and testing. Established by a charter, the SEWG will conduct recurring meetings to share information, resolve issues, and track status and health of EMP survivability and quantify risk to DAF systems and facilities.

3.3.1. The SEWG is comprised of an O-6/GS-15 from each member organization and will meet as needed to review, endorse, or approve charters, EMP policy and guidance, and other products created by the action-officer level EMP Working Group (EWG).

3.3.1.1. AFGSC/A3/6, SF COO/S10 and AF/A10 will tri-chair the SEWG at the O-6/GS-15 level. AFGSC/A3/6 EMP Division Chief will serve as the SEWG Secretariat.

3.3.1.2. Member organizations include, but are not limited to, Headquarters Air Force Directorates, MAJCOMs, and FLDCOMs.

3.3.2. The SEWG is ultimately answerable to the AFGSC/CC as the single accountable officer to the CSAF and SecAF for all aspects of the USAF nuclear mission (specifically those forces in support of USSTRATCOM; however, DAF support to other combatant commands is also addressed by this body). Similarly, the SEWG is answerable to CSO and SecAF through SF/COO for all aspects of the USSF mission within its purview.

3.3.3. To support the SEWG, AFGSC/A3/6 will designate a representative to chair the EWG. The EWG will be comprised of action officers of SEWG member organizations, and its primary role is to assist Lead MAJCOMs/FLDCOMs (systems) or owning MAJCOMs/FLDCOMs (facilities) in determining the impact and mitigation measures for identified EMP survivability shortfalls.

3.3.4. SEWG Members may request SEWG assistance in prioritizing EMP Survivability Program resource requirements and test priorities, to include establishing a standard methodology for each.

3.3.5. Issues may be forwarded or updates provided as requested by other nuclear enterprise activities.

3.3.6. The SEWG, supported by the AFGSC staff, is responsible for producing the annual USAF EMP Survivability and Surveillance Program Report identifying the health and status of EMP survivability for major systems and facilities, EMP test compliance, and a prioritization of shortfall mitigation requirements.

3.3.7. Once SF COO/S10 completes standing up the USSF EMP Program, the DAF SEWG will split into two separate governing bodies, USAF SEWG and USSF SEWG. When the split occurs, USAF members do not have voting privileges in the USSF SEWG, and USSF members do not have voting privilege in USAF SEWG.

3.4. Reporting. Specific procedures will be developed in the SEWG Charter; OPR for the charter is the SEWG Secretariat, AFGSC/A3C.

3.5. Mission Assurance Working Group (MAWG). In addition to the SEWG, AFGSC can also utilize the MAWG. The MAWG is useful to integrate and synchronize efforts across multiple Mission Assurance-Related Programs/Activities.

Chapter 4

REQUIREMENTS

4.1. Overview.

4.1.1. EMP survivability is a needed hardness consideration for new systems and facilities. Requirements development processes must consider if protection is required for the individual system itself or if that system will have dependencies on other features to provide the protection required. When EMP protection is dependent on the environment, such as where a mission is housed (e.g., in a facility), the requirements development process must address protection from the EMP and planning for recovery of commercial power and other support services.

4.1.2. NC3 systems and interfaced facilities require EMP hardening to the limits required for mission recovery as established by USSTRATCOM. USSTRATCOM tests NC3 facilities and supporting infrastructure on a recurring basis. Other C4I sites requiring EMP protection are addressed in CJCSI 3222.01C, *Chairman of the Joint Chiefs of Staff Requirements for Electromagnetic Pulse Protection of Critical Nodes and Systems,* and CJCSI 6810.01C, *Critical Nuclear Command, Control, and Communications Systems and Facilities.* There is no clear, structured process to define EMP hardening requirements for other support elements.

4.1.3. Mission owners should clearly articulate what parts of their mission must be protected and must articulate the speed of recovery after an event (e.g., when commercial power will be restored) so that an appropriate protection system can be configured and managed.

4.1.4. Fielded systems, equipment facilities, and infrastructure require resourced EMP survivability testing activities to sustain originally designed EMP survivability capabilities. EMP survivability testing at the component, system, and threat levels provides feedback on the ability to operate in and through an EMP environment.

4.1.5. The PMO will sustain and test required system EMP survivability capabilities, with assistance from Lead Commands, and provide necessary test assets (such as aircraft) for sustainment. **(T-0)** Reference “System EMP Requirement” in **Terms**.

4.1.6. Owning MAJCOMs, FLDCOMs, and Mission Owners will sustain EMP hardening of existing facilities and supporting infrastructure, and mission owners will sustain EMP hardening of mission systems and equipment when housed inside a facility. **(T-1)**

4.2. Systems and Aircraft. For the purposes of this DAFI, systems include weapons systems, space systems, communications and cyber systems, command and control systems, sensors, munitions, planning systems, and any other non-facility capability that requires EMP hardening for mission assurance.

4.2.1. New Systems. For the purposes of this DAFI, new systems are systems as defined above that have not reached Full Operating Capability. The requirements definition process such as JCIDS Manual establishes the need for EMP hardening for new systems.

4.2.2. Existing Systems. For the purposes of this DAFI, existing systems are systems as defined above that have reached Full Operational Capability.

4.2.3. **Testing.**

4.2.3.1. The PMO will direct development of a test and evaluation strategy to verify new system performance, specifications, acceptance tests, and verification methods in accordance with system requirements documents and applicable MIL-STDs. Additionally, the PMO will develop EMP survivability testing requirements to ensure fielded system hardness is sustained. **(T-1)**

4.2.3.2. Lead MAJCOM/FLDCOM must waive test non-compliance (i.e., not testing) in accordance with **paragraph 5.1** or approve a CSAF or CSO Plan of Actions and Milestones. Alternately, the organization establishing the EMP survivability requirement, if below MAJCOM/FLDCOM, can also waive the requirement. **(T-1)**

4.2.3.3. The PMO, in coordination with AFGSC, plans and programs threat-level testing in accordance with the EMP Test Strategy, coordinated and developed by AFNWC. **(T-1)**

4.2.3.4. The applicable system program will resource all other testing.

4.2.3.5. The Lead Command is responsible for any new EMP hardening or EMP test requirements on existing systems. **(T-1)**

4.2.4. **Unmitigated EMP Survivability Shortfalls.**

4.2.4.1. The PMO will correct any unmitigated shortfall in meeting an established EMP survivability requirement or request a waiver from the Lead Command. (**T-2)**

4.2.4.2. The Lead Command will accept risk for any unmitigated shortfall, not the PMO.

4.2.5. The Lead Command will fund all EMP survivability testing to include threat-level testing through the acquisition process. **(T-2)**

4.2.6. The PMO will coordinate threat-level testing requirements and results of testing and evaluation with AFGSC and AFNWC, as required for new systems. **(T-2)** Once the system is designated as Mission Critical, roles and responsibilities for existing systems apply.

4.2.7. Lead Command will report waived EMP survivability compliance or EMP test compliance to AFGSC and AFNWC for inclusion in the annual DAF EMP Survivability Program Report. **(T-2)**

4.2.8. If requirements documents do not clearly articulate EMP survivability requirements for fielded systems/facilities, the Lead Command will identify the needed requirements to the program office and will work to ensure necessary upgrades/modifications are made to existing systems/facilities unless waived or until replaced by newer capabilities incorporating the required hardness parameters.

4.3. Facilities and Infrastructure.

4.3.1. For the purposes of this DAFI, EMP protection for mission-critical systems, where protection is provided by the environment they are housed in, will be provided by a layered system of protection based on the required speed of recovery after an event. The mission owner bears responsibility for identifying and hardening equipment or facilities based on the length of time the required equipment can be non-operational. The EMP protected environment may be provided by an enclosure installed inside a facility or may be provided by EMP protection measures installed as part of the facility itself. Also reference Facilities and Infrastructure EMP Requirement in **Terms**.

4.3.1.1. For those systems that must be up and running within a few minutes or up to 1 day after an event, it is the responsibility of the mission owner to assure those protections are in place (e.g., shielding, backup power, etc.). **(T-1)**

4.3.1.2. Recovery of commercial power and other support services that will take longer than a day to recover, as well as facility and infrastructure requirements that may be required beyond those required for recovery within 1 day, will be incorporated into new facilities by the mission owners. This includes new construction or renovation of an existing facility. See DAFI 32-1020, *Planning and Programming Built Infrastructure Projects*, for further guidance.

4.3.2. For the purposes of this DAFI, existing facilities are defined as those facilities accepted by the installation and placed in service to be used for their designated purpose. See DAFI 32-9005, *Real Property Accountability*, for additional guidance. Minor facility refurbishment or renovation (below MILCON levels of effort) are accommodated within the existing facility definition; however, the owning MAJCOM/FLDCOM will resource any modifications or changes in EMP hardening features to their installation. **(T-1)**

4.3.3. The requirement for EMP hardening in support of the NC3 mission set is established by USSTRATCOM and the owning Combatant Command, Lead MAJCOM/FLDCOM (e.g., mission owners) for all other mission sets. If the requirement impacts construction activities, those requirements must be established prior to authorization/approval of construction. . EMP hardening non-compliance must be waived by USSTRATCOM or the owning MAJCOM/FLDCOM, as applicable. **(T-0)**

4.3.4. Mission owners will integrate EMP survivability requirements into their mission set equipment requirements and will determine enduring testing requirements to maintain certification. (**T-0)** Coordinate with AFNWC for technical expertise, as required.

4.3.5. AFIMSC, in partnership with the owning MAJCOM/FLDCOM and mission owners, will integrate EMP survivability requirements (guidance provided in MIL-STDs in **Attachment 1**) into environment designs when required and determine enduring testing requirements to maintain certification for real property elements. **(T-0)** Coordinate with AFNWC for technical expertise, as required.

4.3.6. Mission Owners coordinate with USSTRATCOM for real property listed facility and infrastructure testing and certification of NC3 facilities. For other real property listed facilities, the mission owner, in partnership with the owning MAJCOM/FLDCOM, will execute necessary EMP survivability testing and certification. Non-compliance of testing requirements must be waived by USSTRATCOM, the owning MAJCOM/FLDCOM, as required. **(T-0)**

4.3.7. The owning MAJCOM/FLDCOM is responsible for resourcing critical facilities and infrastructure to meet EMP survivability and testing requirements.

4.3.8. The owning MAJCOM/FLDCOM will report waived EMP survivability compliance or EMP test compliance to AFGSC and AFNWC for inclusion in the annual DAF EMP survivability and Surveillance Program report. **(T-0)**

4.3.9. An existing facility being repurposed as a NC3 facility will comply with USSTRATCOM hardening requirements and will be subjected to USSTRATCOM testing. **(T-0)** If USSTRATCOM does not require hardening requirements for specific facilities, the owning MAJCOM/FLDCOM is the authority for defining EMP survivability requirements for those existing facilities. EMP hardening noncompliance must be waived by USSTRATCOM, the owning MAJCOM/FLDCOM, as applicable. **(T-0)**

4.3.10. Owning MAJCOM/FLDCOMs are responsible for:

4.3.10.1. Funding sustainment of existing EMP survivability requirements for real property listed components and ensuring necessary periodic testing is accomplished. **(T-1)**

4.3.10.2. Resourcing the mitigation of shortfalls caused by new MIL-STDs (See **Attachment 1**) or new EMP hardening requirements. **(T-1)**

4.3.11. The owning MAJCOM/FLDCOM is responsible for resourcing all desired facility modifications that impact EMP survivability and testing requirements, and for resourcing the mitigation of any EMP survivability shortfalls caused by facility modifications. **(T-1)**

4.3.11.1. The owning MAJCOM/FLDCOM must notify the Base Civil Engineer organization of planned facility modifications to determine the potential impact to complete and usable functionality of real property and partner with owning MAJCOM/FLDCOM responsible for EMP survivability and testing. **(T-1)**

4.3.11.2. The owning MAJCOM/FLDCOM collaborates with the program office on facility requirements. **(T-1)**

4.3.12. The owning MAJCOM/FLDCOM is responsible for coordinating third party interaction with Base Civil Engineering organization/AFIMSC such as non-DAF tenants and host nations. For non-DAF owned facilities, the using MAJCOM/FLDCOM is responsible for coordinating third-party facility owner interaction with AFIMSC.

4.4. Types of Tests.

4.4.1. Injection Test (also called direct drive test). This is the preferred approach for fixed ground-based facilities and is used most extensively for tests of hardened C3 facilities, missile silos, etc. This simulates the effects of the currents and/or voltages induced by high-altitude EMP (HEMP) on cables by artificially injecting current pulses onto/into equipment cables and wires. Injection tests are particularly well suited to the evaluation of interior equipment that is not directly exposed to HEMP and amenable to in situ testing. Also utilized in aircraft active systems testing is to demonstrate that the aircraft design margin (DM) has been achieved and to evaluate conductive penetrations of the EM barrier that may include current injection onto radio frequency (RF) antenna core wires, control wires, trailing wire antenna (TWA), in-flight refueling boom, and ground alert mode wiring to verify systems survive and can still perform their mission critical functions.

4.4.2. Free-Field Test: Also known as threat-level field illumination is the preferred approach for mobile systems unattached to long line networks (e.g., aircraft, ships, and transportable C3I vans). This is used to expose equipment, such as missiles, aircraft, vehicles, radar antenna, and EM radiation equipment, to HEMP. Most free-field HEMP testing is performed with either a broadcast simulator or a bounded wave EMP simulator. For aircraft this includes active systems testing which consist of illuminating the complete aircraft at HEMP threat level with the aircraft in a power-on state to confirm MCS continue to operate satisfactory. Also includes passive system test which consists of illuminating the complete aircraft at HEMP threat level with the aircraft in a simulated power-on state to measure currents induced at stress/strength interface locations to validate design margins and identify test points for direct for drive testing.

4.4.3. Continuous Wave Immersion: This is similar to free-field testing, except the signal intensity is significantly reduced to very low signal levels, and the equipment is subjected to a number of sinusoidal swept frequencies for a significantly longer period of time to captured all measured current induced interfaces and/or enclosure test points. Provides threat relatable data at test points but does not demonstrate immunity or threat survivability. This test is often performed as a risk-reduction test for a threat-level test, or as part of a HM/HS plan. Additional testing required at threat level to determine survivability of MCS.

4.4.4. Immunity testing: Testing of a component, mission critical system, or system to verify its ability to perform without degradation of performance in the presence of a simulated HEMP type stress IAW MIL-STD-461 CS-116 and RS-105 susceptibility tests.

Chapter 5

RISK DECISION

5.1. Waiver Process (applicable to USAF only; USSF does not have a waiver process).

5.1.1. A PMO seeking either an EMP requirement or testing waiver will submit a memorandum to the EWG thirty (30) days prior to submitting the waiver to the lead MAJCOM/FLDCOM (or appropriate mission owner). The memorandum will include justification for the waiver and the SEWG risk assessment. **(T-1)**

5.1.2. Lead Command/Mission Owner will either approve or deny the waiver request and then send AFGSC a memorandum, courtesy copy to the SEWG and requesting PMO, stating the risk that is being accepted by waiving the requirement. **(T-1)**

5.1.3. Lead Command will annually review waivers granted for longer than one (1) year and validate whether they are still required. If the waiver is extended, the Lead Command will then send a memorandum to AFGSC, courtesy copy the SEWG and the requesting PMO, stating the risk that is being accepted if it is extended. **(T-1)**

5.1.4. For facilities and infrastructure, mission owner maintains full responsibility for implementing a requirement or waiving the EMP protection requirement, as their mission needs require it. **(T-1)**

5.2. Waiver Cancellation Process. The PMO will submit a memorandum to the SEWG stating which waiver/waivers it intends to cancel along with documentation showing that the waived requirement(s) are now being met. **(T-1)**

ANDREW J. GEBARA, Lt Gen, USAF
DCS, Strategic Deterrence & Nuclear Integration

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DoDD 3020.40, *Mission Assurance,* 29 November 2016

DoDD 8260.05, *Support for Strategic Analysis (SSA),* 7 July 2011

DoDI 5000.89_DAFI 99-103, *Capabilities-Based Test and Evaluation,* 15 March 2022

DoDI 3020.45-V1, *Mission Assurance Construct,* 14 August 2018

MIL-STD-188-125-1A, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Ground-Based C4I Facilities Performing Critical, Time-Urgent Missions: Part 1 - Fixed Facilities,* 1 July 2021

MIL-STD-188-125-2, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Ground-Based C4I Facilities Performing Critical, Time-Urgent Missions: Part 2 - Transportable Facilities,* 7 April 2005

MIL-STD-461G, *Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment,* 11 December 2015

MIL-STD-464D, *Electromagnetic Environmental Effects Requirements for Systems,* 24 December 2020

MIL-STD-1916, *DOD Preferred Methods for Acceptance of Product,* 5 June 2014

MIL-STD-1766C (USAF*), Nuclear Hardness and Survivability Program Requirements ICBM Weapon Systems,* 17 January 2017

MIL-STD-2169D, *(U) High-Altitude Electromagnetic Pulse (HEMP) Environment (SECRET),* 30 March 2021

MIL-STD-3023, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Military Aircraft,* 21 November 2011

MIL-STD-3053, *(U) Satellite System Natural and Nuclear Environments Standard (SSNS-E) (SECRET//RESTRICTED DATA-N)*

MIL-HDBK-423, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Fixed and Transportable Ground-Based C4I Facilities, Volume I Fixed Facilities,* 19 November 2019

MIL-HDBK-516C, *Airworthiness Certification Criteria,* 7 October 2019

MIL-HDBK-2069, *Aircraft Survivability,* 30 August 2019

CJCSI 3222.01C, *Chairman of the Joint Chiefs of Staff Requirements for Electromagnetic Pulse Protection of Critical Nodes and Systems,* 8 August 2019

CJCSI 3810.01G, *Meteorological and Oceanographic Support,* 31 March 2023

CJCSI 6810.01C, *Critical Nuclear Command, Control, and Communications Systems and Facilities,* 29 August 2019

JCIDS Manual

NC3 Directive (ND) 501-02, *Integrated Chemical, Biological, Radiological, and Nuclear (CBRN) Survivability and Endurability Certification & Assessment Program (ICSECAP),* 31 October 2021

DAFPD 10-9, *Lead Command/Lead Agent Designation and Responsibilities for United States Air Force Weapon Systems, Non-Weapon Systems, and Activities,* 25 May 2021

DAFPD 10-26, *Countering Weapons of Mass Destruction,* 19 December 2024

DAFI 10-208, *Continuity of Operations (COOP) Program*, 12 May 2023

DAFI 20-101_63-101, *Integrated Lifecycle Management,* 16 February 2024

DAFI 32-1020, *Planning and Programming Built Infrastructure Projects*, 18 December 2019

DAFI 32-9005, *Real Property Accountability*, 14 September 2022

DAFI 90-302, *The Inspection System of the Department of the Air Force,* 03 September 2024

AFPD 10-6, *Capability Requirements Development*, 6 November 2013

AFPD 10-24, *Mission Assurance,* 5 November 2019

AFPD 13-5, *Air Force Nuclear Mission,* 17 July 2018

AFI 10-201, *Force Readiness Reporting*, 27 April 2023

AFI 10-601, *Operational Capability Requirements Development and Validation,* 27 April 2021

AFI 10-2607, *Air Force Chemical, Biological, Radiological, and Nuclear Survivability,* 9 August 2018

AFI 13-500, *Air Force Nuclear Mission Responsibilities*, 14 November 2018

AFI 33-322, *Records Management and Information Governance Program,* 23 March 2020

SPFI 10-201, *Force Readiness Reporting*, 14 February 2024

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*.

Abbreviations and Acronyms

AETC—Air Education Training Command

AFGSC—Air Force Global Strike Command

AFI—Air Force Instruction

AFIMSC—Air Force Installation and Mission Support Center

AFMC—Air Force Materiel Command

AFNWC—Air Force Nuclear Weapons Center

AFPD—Air Force Policy Directive

BSA—Defense Threat Reduction Agency Balanced Survivability Assessments

CBRN—Chemical Biological Radiological Nuclear

C4I—Command, Control, Communications, Computers, and Intelligence

CJCSI—Chairman of the Joint Chiefs of Staff Instruction

COO—Chief Operations Officer

CSAF—Chief of Staff of the Air Force

CSO—Chief of Space Operations

CSRO—Chief Strategy and Resourcing Officer

CTIO—Chief Technology and Innovation Officer

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DEMP—Dispersed Electromagnetic Pulse

DoD—Department of Defense

DoDD—Department of Defense Directive

DoDI—Department of Defense Instruction

DRU—Direct Reporting Unit

EEPMO—Electromagnetic Pulse Effects Program Management Office

EMP—Electromagnetic Pulse

ETPL—Electromagnetic Pulse Test Priority List

EWG—Electromagnetic Pulse Working Group

FLDCOM—Field command

GMD—Geomagnetic Disturbance

HEMP—High Altitude Electromagnetic Pulse

HPMO—HEMP Program Management Office

HM/HS—Hardness Maintenance & Hardness Surveillance

ITW/AA—Integrated Tactical Warning/Attack Assessment

JCIDS—Joint Capabilities Integration and Development System

MAJCOM—Major command

MAWG—Mission Assurance Working Group

MILCON—Military Construction

MIL-STD—Military Standard

NC3—Nuclear Command, Control, & Communications

NLCC—National Leadership Command Capabilities

OPR—Office of Primary Responsibility

PME—Professional Military Education

PMO—Program Management Office

S/E—Survivable/Endurable

SecAF—Secretary of the Air Force

SEWG—Senior Electromagnetic Pulse Working Group

SGEMP—System Generated Electromagnetic Pulse

SpOC—Space Operations Command

STARCOM—Space Training and Readiness Command

USAF—United States Air Force

USSF—United States Space Force

USSTRATCOM—United States Strategic Command

Office Symbols

AF/A2—Deputy Chief of Staff for Intelligence

AF/A3—Deputy Chief of Staff for Operations

AF/A4—Deputy Chief of Staff for Logistics, Engineering, and Force Protection

AF/A5/7—Deputy Chief of Staff for Air Force Futures

AF/A6—Deputy Chief of Staff for Warfighter Communications and Cyber Systems

AF/A8—Deputy Chief of Staff for Plans and Programs

AF/A10—Deputy Chief of Staff for Strategic Deterrence and Nuclear Integration

AF/RE—Chief of Air Force Reserve

AF/SG—Air Force Surgeon General

AF/TE—Air Force Director of Test and Evaluation

SAF/AQ—Assistant Secretary of the Air Force for Acquisition, Technology, and Logistics

SAF/FM—Assistant Secretary of the Air Force for Financial Management and Comptroller

SAF/IA—Deputy Under Secretary of the Air Force for International Affairs

SAF/IE—Assistant Secretary of the Air Force for Installations, Environment, and Energy

SAF/IG—Department of the Air Force Inspector General

SAF/SA—Director of Studies and Analysis

SAF/SQ—Assistant Secretary of the Air Force for Space Acquisition and Integration

USSF/S3/4/6/7/10—Deputy Chief of Space Operations for Operations, Cyber, and Nuclear

USSF/S5/8—Deputy Chief of Space Operations for Strategy, Plans, Programs, and Requirements

Terms

Dispersed Electromagnetic Pulse (DEMP)—Refers to the HEMP environment propagated through the ionosphere that is experienced by a space-based system.

Electromagnetic Pulse (EMP)—A strong burst of electromagnetic radiation caused by a nuclear explosion, energy weapon, or by natural phenomenon (i.e., solar-induced space weather), that may couple with electrical or electronic systems to produce damaging current and voltage surges.

Electromagnetic Pulse Hardness—The extent that a system or facility is protected, tested, and certified to operate in or through a HEMP event in accordance with applicable Military Standards (MIL-STD). Many concepts of EMP survivability may be achieved through other requirements, such as the MIL-STD-461G, Electromagnetic Compatibility, among others.

Facilities and Infrastructure EMP Requirement—For the purposes of this DAFI, the identification process for mission-critical systems will be accomplished using guidance provided in AFI 10-2607.

Geomagnetic Disturbance (GMD)—Also known as a magnetic storm, a GMD is a temporary disturbance of the Earth's magnetosphere caused by a solar wind shockwave.

Hardness Maintenance/Hardness Surveillance (HM/HS)—With respect to EMP survivability, HM/HS is the collected activities to sustain designed EMP survivability protection through sustainment, and to verify EMP survivability performance through periodic assessment.

Hardness Maintenance—The collected activities by the using command and depot-level repair facilities to maintain and preserve the hardness of fielded weapon systems throughout operational life, especially with respect to spares procurement and repair actions as well as activities such as corrosion prevention, EMP shielding maintenance, and configuration control in accordance with Technical Orders (TO) and associated maintenance guidance.

Hardness Surveillance—Periodic hardness tests conducted at the component, sub-system, system, and threat-level environments with the purpose of identifying on a timely manner any degradations that reduce the EMP survivability of fielded systems and facilities. Such degradations may be due to aging, environmental effects, continuous operation, maintenance actions, or system modifications.

Lead Command—The command that serves as the using command's interface with the program manager for a system as prescribed by AFPD 10-9, *Lead Command/Lead Agent Designation and Responsibilities for United States Air Force Weapon Systems, Non-Weapon Systems, and Activities*. Lead Command designation is not exclusive to MAJCOMs and FLDCOMs; Field Operating Agencies and DRUs may also be designated.

Mission Owner—The DoD Component or subordinate unit having responsibility for the execution of all or part of a mission assigned by statute or the Secretary of Defense. (DoDD 3020.40)

Mission Critical System—A system whose operational effectiveness and operational suitability are essential to successful mission completion or to aggregate residual combat capability. If this system fails, the mission likely cannot be completed. Such a system can be an auxiliary or supporting system, as well as a primary mission system. (DoDI 3150.09)

Risk Acceptance—The informed decision by the Lead MAJCOM/FLDCOM for assigned systems, or owning MAJCOM/FLDCOM for facilities, to intentionally forego EMP HM/HS on new or existing capabilities.

System EMP Requirement—For the purposes of this DAFI, a system includes weapon systems, communications and cyber systems, command and control systems, sensors, munitions, planning systems, and any other non-facility capability that requires EMP hardening in accordance with MIL-STDs.

System Generated Electromagnetic Pulse (SGEMP)—Refers to the EMP environment caused by nuclear weapon produced x-ray interaction with a space-based system.

Testing—For the purposes of this DAFI, testing includes pre- and post-testing and analysis processes.